Assignment #1: Write an assemble codes to accept string from keyboard and display it on the screen in reverse order on position (10,30) with foreground color green and background color white

Solution:

```
ORG 100H

            MOV AH,06H
            XOR AL,AL
            XOR CX,CX
            MOV DX,184FH
            MOV BH,0F2H
            INT 10H

            MOV AH,09H
            LEA DX,MESSAGE
            INT 21H

            LEA SI,SEN
            MOV AH,0AH
            MOV DX,SI
            INT 21H

            MOV CL,SEN
            ADD CL,1
            ADD SI,2
Laith1:     INC SI
            CMP [SI],"$"
            JNE Laith1

            DEC SI
            LEA DI,REV
Laith2:     MOV AL,[SI]
            MOV [DI],AL
            DEC SI
            INC DI
            LOOP Laith2

            MOV AH,09H
            LEA DX,REV
            INT 21H

HLT

    MESSAGE DB "INPUT THE STRING [ MAXIMUM 20 CHARS ] : ",0AH,0AH,0DH,"$"
    SEN DB 21,?,21 DUP("$")
    REV DW 21 DUP("$")
```

ليث، عمرو

Assignment #2: Write an assembly code to accept X, Y from key board and perform the following operations:

- Call procedure to convert X from ASCII to binary
- Call procedure to convert Y to binary.
- Add the binary value of X and the binary value of Y storing the result into ZBINAR
- Call a procedure to convert ZBINARY to Z ASCII
- Call a procedure to print ZASCII on the screen

```
ORG 100H

        MOV AH,09H
        LEA DX,X
        INT 21H

        MOV AH,0AH
        LEA DX,X_INPUT
        INT 21H

        CALL ATBX
        MOV SEN[0],AL

        MOV AH,09H
        LEA DX,Y
        INT 21H

        MOV AH,0AH
        LEA DX,Y_INPUT
        INT 21H

        CALL ATBY
        MOV SEN[1],AL

        CALL XSUMY

        CALL BTA

        CALL PRINT


HLT

    ;*****************************

    X DB "ENTER THE VALUE OF VARIABLE X : ",0AH,0AH,0DH,"$"
    X_INPUT DB 3,?,3 DUP(?)
    Y DB 0AH,0DH,"ENTER THE VALUE OF VARIABLE Y : ",0AH,0AH,0DH,"$"
    Y_INPUT DB 3,?,3 DUP(?)
    SEN DB 2 DUP(?)
    Z_RESULT DB 0AH,0DH,"THE RESULT IS : ","$"
    Z_OUTPUT 2 DUP(?),"$"

        ;*****************************
```

ليث، عمرو

```
ATBX PROC
MOV SI,3
MOV BX,OFFSET X_INPUT
AND [BX+SI],0FH
MOV DL,[BX+SI]
DEC SI
AND [BX+SI],0FH
MOV AL,[BX+SI]
MOV BL,10
MUL BL
ADD AL,DL
RET
ATBX ENDP
;*****************************
ATBY PROC
MOV SI,3
MOV BX,OFFSET Y_INPUT
AND [BX+SI],0FH
MOV DL,[BX+SI]
DEC SI
AND [BX+SI],0FH
MOV AL,[BX+SI]
MOV BL,10
MUL BL
ADD AL,DL
RET
ATBY ENDP
;*****************************
BTA PROC
MOV BL,10
MOV AL,SEN[0]
DIV BL
ADD AH,30H
ADD AL,30H
MOV Z_OUTPUT[1],AH
MOV Z_OUTPUT[0],AL
RET
BTA ENDP
;*****************************
XSUMY PROC
MOV BL,SEN[1]
ADD SEN[0],BL
MOV AL,SEN[0]
RET
XSUMY ENDP
;*****************************
PRINT PROC
MOV AH,09H
LEA DX,Z_RESULT
INT 21H
MOV DX,OFFSET Z_OUTPUT
MOV AH,09H
INT 21H
RET
PRINT ENDP
```

ليث، عمرو